

Engineering Project Report

Simple DC-DC Boost Converter Circuit Using NE555 Timer IC and IRFZ44N MOSFET

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Abstract

This engineering report presents the design, mathematical formulation, circuit implementation, and power output testing of a non-isolated DC-DC Boost Converter using an NE555 timer IC. The 555 IC functions in astable mode as a Pulse Width Modulation (PWM) oscillator driving an IRFZ44N N-Channel Power MOSFET switch. Energy stored in a $100\text{ }\mu\text{H}$ toroidal inductor during the MOSFET conduction phase is transferred to a $100\text{ }\mu\text{F}$ output filter capacitor through a 1N5822 Schottky diode during the off phase. Powered by a single $3.4\text{ V} - 3.7\text{ V}$ Li-ion battery, the converter successfully steps up the voltage to $7.5\text{ V} - 12\text{ V}$ DC, delivering output power levels up to 30 Watts (3.2 A short-circuit capability). Complete mathematical derivations, duty cycle control relations, component selection rules, and closed-loop feedback upgrade recommendations are detailed.

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1 Introduction & SMPS Concepts

Switch-Mode Power Supplies (SMPS) offer high power conversion efficiency ($\geq 85\%$) compared to linear regulators by operating switching components (MOSFETs) strictly in saturation (ON) or cutoff (OFF) modes.

A ****Boost Converter**** (Step-Up Converter) increases DC voltage while decreasing output current proportionally, satisfying conservation of power ($P_{in} \approx P_{out}$).

2 Bill of Materials

Table 1: Bill of Materials for 555-Based Boost Converter

S.No.	Component	Qty	Part Number	Circuit Function
1	PWM Timer IC	1	NE555N	High-frequency PWM oscillator
2	Power Switch	1	IRFZ44N MOSFET	55 V, 49 A, 17.5 mΩ N-Channel switch
3	Storage Inductor	1	100 μ H Toroidal	Inductive energy storage element (≥ 3 A)
4	Rectifier Diode	1	1N5822 Schottky	Fast-recovery output diode (40 V, 3 A)
5	Steering Diodes	2	1N4001 / 1N4148	Astable duty-cycle control network
6	Filter Capacitor	1	100 μ F, 35V	High-voltage output DC smoothing
7	Timing Capacitors	2	1 nF, 100 nF	555 timing (C_1) and control filter (C_2)
8	Potentiometer	1	50 kΩ Trimpot	PWM duty cycle (D) adjustment
9	Connector Terminals	2	2-Pin Screw Blocks	Input and output DC connections

3 Mathematical Analysis & Operational Equations

3.1 Two-Stage Switching Operation

- Stage 1: MOSFET ON (T_{on} Phase):** Current flows from $V_{in} \rightarrow$ Inductor $\rightarrow$ MOSFET $\rightarrow$ GND. Energy is stored in the magnetic field ($E_L = \frac{1}{2}LI^2$). The Schottky diode is reverse-biased.
- Stage 2: MOSFET OFF (T_{off} Phase):** The MOSFET turns OFF. The inductor opposes current drop by producing a flyback voltage ($V_L = L\frac{di}{dt}$) in series with V_{in} , forward-biasing the Schottky diode to charge the output capacitor to V_{out} .

3.2 Duty Cycle & Voltage Step-Up Ratio

The switching period is $T = T_{on} + T_{off}$. The duty cycle D is:

$$D = \frac{T_{on}}{T} \quad (1)$$

Applying Kirchhoff's Voltage Law across one complete switching cycle yields the steady-state boost ratio:

$$V_{out} = \frac{V_{in}}{1 - D} \quad (2)$$

3.3 Sample Boost Calculation

For an input $V_{\text{in}} = 3.7 \text{ V}$ and a duty cycle $D = 0.65$:

$$V_{\text{out}} = \frac{3.7}{1 - 0.65} = \frac{3.7}{0.35} \approx 10.57 \text{ V}$$

3.4 Inductor Current Ripple Formula

The peak-to-peak inductor ripple current (ΔI_L) is given by:

$$\Delta I_L = \frac{V_{\text{in}} \cdot D}{f \cdot L} \quad (3)$$

where f is 555 switching frequency ($\approx 25 \text{ kHz}$) and $L = 100 \mu\text{H}$.

4 Performance Evaluation & Experimental Results

- **Input Source:** 3.4 V – 3.7 V Li-ion Cell (2000 mAh).
- **Measured Output Voltage:** 7.5 V – 12.0 V DC (Adjustable via 50 k Ω pot).
- **Peak Short-Circuit Current:** 3.2 Amperes ($\sim 30 \text{ W}$ peak handling capacity).
- **Note on Construction:** High peak currents ($> 3 \text{ A}$) necessitate assembly on a soldered Perfboard/PCB to avoid breadboard trace melting.

5 Closed-Loop Feedback Upgrade Recommendation

This basic circuit runs in ****open-loop mode**** without feedback. Connecting a heavy load causes voltage sag. For a regulated 12 V output, add a feedback resistive divider connected to a microcontroller ADC (or TL431 op-amp) to dynamically adjust 555 PWM duty cycle (D) under varying load currents.

6 Conclusion

The NE555 + IRFZ44N DC-DC Boost Converter effectively steps up low DC voltages (3.7 V $\rightarrow$ 12 V) with high power density ($\sim 30 \text{ W}$), illustrating fundamental SMPS switching principles.